

Def. 1 (Divisibility)
Let $a, b \in \mathbb{Z}$ with $a \neq 0$. We say a divides b if $\exists c \in \mathbb{Z}$ such that $b = ac$. Write $a \mid b$; otherwise $a \nmid b$.

Thm. 2 (Division Algorithm)
If $a, b \in \mathbb{Z} \setminus \{0\}$, then there exist unique $q, r \in \mathbb{Z}$ such that

$$a = qb + r, \quad 0 \leq r < |b|.$$

Rem. 3
If $a = qb + r$ with $0 \leq r < |b|$, then the common divisors of a, b are exactly the common divisors of b, r . In particular, $(a, b) = (b, r)$.

Def. 4 (Greatest Common Divisor (g.c.d.))
For $a, b \in \mathbb{Z} \setminus \{0\}$, the g.c.d. of a and b is the unique positive integer $(a, b) = d$ such that:

- $d \mid a$ and $d \mid b$;
- if $e \mid a$ and $e \mid b$, then $e \mid d$.

If $(a, b) = 1$, we say a, b are relatively prime.

Def. 5 (Least Common Multiple (l.c.m.))
For $a, b \in \mathbb{Z} \setminus \{0\}$, the l.c.m. of a and b is the unique positive integer $[a, b] = l$ such that:

- $a \mid l$ and $b \mid l$;
- if $a \mid m$ and $b \mid m$, then $l \mid m$.

Prop. 6
For $a, b \in \mathbb{Z} \setminus \{0\}$,

$$(a, b) [a, b] = ab.$$

Thm. 7 (Bézout Identity)
If $a, b \in \mathbb{Z} \setminus \{0\}$, then there exist $x, y \in \mathbb{Z}$ such that

$$(a, b) = ax + by.$$

Prop. 8
If $(a, b) = 1$, then $\exists x, y \in \mathbb{Z}$ such that

$$ax + by = 1.$$

Prop. 9 (Key Divisibility Property)
If $a \mid bc$ and $(a, b) = 1$, then $a \mid c$.

Def. 10 (Prime)
A prime is an element $p \in \mathbb{Z}^+$ such that $p > 1$ and its only positive divisors are 1 and p . A positive integer which is not prime is called composite.

Thm. 11 (Fundamental Theorem of Arithmetic)
If $n \in \mathbb{Z}$ and $n > 1$, then n can be factored uniquely as

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s},$$

where $p_1, \dots, p_s$ are distinct primes and $\alpha_1, \dots, \alpha_s \in \mathbb{Z}^+$.

Prop. 12 (g.c.d. via Prime Powers)
If

$$a = p_1^{\alpha_1} \cdots p_s^{\alpha_s}, \quad b = p_1^{\beta_1} \cdots p_s^{\beta_s}$$

(with exponents ≥ 0), then

$$(a, b) = p_1^{\min(\alpha_1, \beta_1)} \cdots p_s^{\min(\alpha_s, \beta_s)}.$$

Def. 13 (Congruence Mod n)
We say a is congruent to b mod n if $n \mid (a - b)$, written

$$a \equiv b \pmod{n}.$$

The congruence class of a mod n is

$$\bar{a} = \{a + kn : k \in \mathbb{Z}\}.$$

There are exactly n distinct classes $\bar{0}, \bar{1}, \dots, \overline{n-1}$. The set of all classes is denoted $\mathbb{Z}/n\mathbb{Z}$.

Def. 14 (Modular Arithmetic)

For $\bar{a}, \bar{b} \in \mathbb{Z}/n\mathbb{Z}$, define

$$\bar{a} + \bar{b} = \overline{a + b}, \quad \bar{a} \cdot \bar{b} = \overline{ab}.$$

Thm. 15
The addition and multiplication defined on $\mathbb{Z}/n\mathbb{Z}$ are well-defined.

Def. 16 (Units Mod n)

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} : \exists \bar{c} \in \mathbb{Z}/n\mathbb{Z} \text{ with } \bar{a} \cdot \bar{c} = \bar{1}\}.$$

Rem. 17

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} : (a, n) = 1\}.$$

Def. 18 (Binary Operation)

A binary operation $\circ$ on a set G is a function $\circ : G \times G \rightarrow G$. For $a, b \in G$ we write $a \circ b$.

Def. 19 (Associative / Commutative)

$\circ$ is associative if $a \circ (b \circ c) = (a \circ b) \circ c$ for all $a, b, c \in G$. It is commutative if $a \circ b = b \circ a$ for all $a, b \in G$.

Def. 20 (Closed Subset)

Let $H \subseteq G$. If $a, b \in H \Rightarrow a \circ b \in H$, then H is closed under $\circ$.

Def. 21 (Group)

A group is an ordered pair $(G, \circ)$ where G is a set and $\circ$ is a binary operation on G such that:

- $\circ$ is associative;
 - $\exists e \in G$ (identity) with $a \circ e = e \circ a = a$ for all $a \in G$;
 - $\forall a \in G \exists a^{-1} \in G$ with $a \circ a^{-1} = a^{-1} \circ a = e$.
- Finite group: G finite. Abelian: $\circ$ commutative.

Thm. 22 (Basic Group Facts)

Let $(G, \circ)$ be a group.

- The identity element is unique.
- Each $a \in G$ has a unique inverse.
- $(a^{-1})^{-1} = a$.
- $(a \circ b)^{-1} = b^{-1} \circ a^{-1}$.
- (Cancellation) If $a \circ c = b \circ c$ or $c \circ a = c \circ b$, then $a = b$.
- The equations $a \circ x = b$ and $y \circ a = b$ have unique solutions in G .

Thm. 23 (Left Identity + Left Inverse Criterion)
An ordered pair $(G, *)$ is a group iff:

- $*$ is associative;
- $\exists e \in G$ such that $e * a = a$ for all $a \in G$ (left identity);
- $\forall a \in G \exists b \in G$ such that $b * a = e$ (left inverse).

Def. 24 (Multiplication Table)

Let $G = \{g_1, \dots, g_n\}$ be finite with $g_1 = e$. The group table is the $n \times n$ matrix with (i, j) entry $g_i g_j$.

Def. 25 (Order of an Element)

For $x \in G$, the order $o(x)$ is the smallest $n \in \mathbb{Z}^+$ such that $x^n = e$ (if it exists).

Thm. 26 (Order Properties)

Let $(G, *)$ be a group and $a \in G$ with $o(a) = n$.

- If $a^m = e$ for some $m \in \mathbb{Z}^+$, then $n \mid m$.
- For every $t \in \mathbb{Z}^+$,

$$o(a^t) = \frac{n}{(t, n)}.$$

In particular, if $(t, n) = 1$, then $o(a^t) = n$.

Def. 27 (Generators)

A subset $S \subseteq G$ is a set of generators of a group G if every element of G can be written as a finite product of elements of S and their inverses. We write $G = \langle S \rangle$.

Def. 28 (Presentation)

A presentation of G is given by generators and relations:

$$G = \langle S \mid R_1, R_2, \dots, R_m \rangle,$$

where each relation is an equality between two finite products of elements of S and their inverses.

Def. 29 (Cyclic Group)

A group G is cyclic if it is generated by one element:

$$G = \langle x \rangle.$$

Examples: $\mathbb{Z}/n\mathbb{Z} = \langle \bar{1} \rangle$, $\mathbb{Z} = \langle 1 \rangle$.

Def. 30 (Quadratic Field)

Let D be square-free. Define

$$\mathbb{Q}[\sqrt{D}] = \{a + b\sqrt{D} : a, b \in \mathbb{Q}\}.$$

Addition and multiplication are

$$(a + b\sqrt{D}) + (x + y\sqrt{D}) = (a + x) + (b + y)\sqrt{D},$$

$$(a + b\sqrt{D})(x + y\sqrt{D}) = (ax + byD) + (ay + bx)\sqrt{D}.$$

Prop. 31

$(\mathbb{Q}[\sqrt{D}], +)$ is a group, and $(\mathbb{Q}[\sqrt{D}]^\times, \cdot)$ is a group.

Def. 32 (Quaternion Group)

$$Q_8 = \{1, -1, i, -i, j, -j, k, -k\},$$

with multiplication determined by:

- $1 \cdot a = a \cdot 1 = a$ for all $a \in Q_8$;
- $(-1) \cdot (-1) = 1$, and $(-1) \cdot a = a \cdot (-1) = -a$;
- $i^2 = j^2 = k^2 = -1$;
- $ij = k, ji = -k$;
- $jk = i, kj = -i$;
- $ki = j, ik = -j$.

Def. 33 (Quaternions)

$$\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}.$$

Elements of $\mathbb{H}$ are called quaternions.

Def. 34 (Dihedral Group)

$$D_{2n} = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle.$$

Rem. 35

In D_{2n} :

- $1, r, \dots, r^{n-1}$ are all distinct (since $o(r) = n$).
- $s \neq r^i$ for any i .
- $sr^i \neq sr^j$ for $0 \leq i \neq j \leq n-1$.

Hence

$$D_{2n} = \{1, r, \dots, r^{n-1}, s, sr, \dots, sr^{n-1}\}.$$

Rem. 36

If $sr^i \in D_{2n}$, then $o(sr^i) = 2$.

Def. 37 (Example: Presentation Y)

$$Y = \langle x, y \mid x^4 = y^3 = 1, xy = y^2x^2 \rangle.$$

Rem. 38

Fact: Among the groups above, D_{2n} ($n \geq 3$) and Q_8 are non-abelian.

Def. 39 (Symmetric Group)

Let $\Omega = \{1, 2, \dots, n\}$. The set of all bijections $\Omega \rightarrow \Omega$ forms a group under composition, denoted

$$(S_\Omega, \circ) = S_n.$$

Rem. 40

Fact: $|S_n| = n!$.

Def. 41 (m -cycle)

The permutation sending

$$a_1 \mapsto a_2 \mapsto \cdots \mapsto a_m \mapsto a_1$$

is denoted $(a_1, a_2, \dots, a_m)$ and is called an m -cycle.

Thm. 42 (Cycle Decomposition)

Every $\sigma \in S_n$ can be written as a product of disjoint cycles:

$$(a_1, \dots, a_{m_1})(a_{m_1+1}, \dots, a_{m_2}) \cdots (a_{m_{k-1}+1}, \dots, a_{m_k})$$

and this decomposition is unique up to reordering of the factors.

Rem. 43

Fact: S_n is non-abelian for all $n \geq 3$.

Rem. 44

Fact: The order of an m -cycle $(a_1, \dots, a_m)$ is m .

Prop. 45 (Order of a Permutation)

The order of $\sigma \in S_n$ is the l.c.m. of the lengths of the cycles in its disjoint cycle decomposition.

Prop. 46 (2-cycle Decomposition)

Any $\sigma \in S_n$ can be written as a product of 2-cycles. In particular, any k -cycle satisfies

$$(i_1, i_2, \dots, i_k) = (i_1, i_k)(i_1, i_{k-1}) \cdots (i_1, i_2).$$

Thm. 47 (Generators of S_n)

$$S_n = \langle (1, 2), (1, 3), \dots, (1, n) \rangle.$$

Moreover, for any 2-cycle (a, b) ,

$$(a, b) = (1, a)(1, b)(1, a).$$

Def. 48 (Even / Odd Permutation)

A permutation $\sigma \in S_n$ is even if it is a product of an even number of 2-cycles; otherwise it is odd.

Rem. 49

Example: An m -cycle $(a_1, \dots, a_m)$ is even iff m is odd, and odd iff m is even.

Def. 50 (Alternating Group)

Let $A_n \subseteq S_n$ be the set of even permutations.

Thm. 51

A_n is a group (the alternating group).

Def. 52 (Subgroup)

Let G be a group. A subset $H \subseteq G$ is a subgroup if $H \neq \emptyset$ and is closed under products and inverses:

$$x, y \in H \Rightarrow x^{-1} \in H, xy \in H.$$

Write $H \leq G$.

Prop. 53 (Subgroup Criterion)

A subset $H \subseteq G$ is a subgroup iff:

- $H \neq \emptyset$,
- for all $x, y \in H$, we have $xy^{-1} \in H$.

Def. 54 (Center)

$$Z(G) = \{g \in G : gx = xg \text{ for all } x \in G\}.$$

Prop. 55

$Z(G) \leq G$.

Prop. 56 (Center of D_{2n} , $n \geq 3$)

$$Z(D_{2n}) = \begin{cases} \{1, r^{n/2}\}, & n \text{ even,} \\ \{1\}, & n \text{ odd.} \end{cases}$$

Def. 57 (Centralizer and Normalizer)

Let $A \subseteq G$ be nonempty. Define

$$C_G(A) = \{g \in G : gag^{-1} = a \text{ for all } a \in A\}.$$

Define $gAg^{-1} = \{gag^{-1} : a \in A\}$ and

$$N_G(A) = \{g \in G : gAg^{-1} = A\}.$$

Prop. 58

For nonempty $A \subseteq G$, both $C_G(A)$ and $N_G(A)$ are subgroups of G .

Def. 59 (Subgroup Generated by A)

Let $A \subseteq G$ be any subset. Define

$$\langle A \rangle = \bigcap_{\substack{A \subseteq H \\ H \leq G}} H = \{a_1^{n_1} a_2^{n_2} \cdots a_r^{n_r} : a_i \in A, n_i \in \mathbb{Z}, r \in \mathbb{N}\}$$

Def. 60 (Cyclic Group)

A group G is cyclic if it can be generated by a single element:

$$G = \langle x \rangle.$$

Examples: $(\mathbb{Z}/n\mathbb{Z}, +) = \langle \bar{1} \rangle$, $(\mathbb{Z}, +) = \langle 1 \rangle$.

Prop. 61

If $G = \langle x \rangle$, then $|G| = o(x)$. More specifically:

- If $|G| = n < \infty$, then $x^n = 1$ and

$$G = \{1, x, x^2, \dots, x^{n-1}\}.$$

- If $|G| = \infty$, then $x^n \neq 1$ for all $n \neq 0$ and $x^a \neq x^b$ for all $a \neq b$ in $\mathbb{Z}$.

Prop. 62
Let G be a group, $x \in G$, and $m, n \in \mathbb{Z}$. If $x^n = 1$ and $x^m = 1$, then

$$x^{(m,n)} = 1.$$

Prop. 63
Let $G = \langle x \rangle$.

- If $o(x) = \infty$, then $G = \langle x^a \rangle$ iff $a = \pm 1$.
- If $o(x) = n < \infty$, then $G = \langle x^a \rangle$ iff $(a, n) = 1$.

Example: $\mathbb{Z}/12\mathbb{Z} = \langle \bar{1} \rangle = \langle \bar{5} \rangle = \langle \bar{7} \rangle = \langle \bar{11} \rangle$.

Thm. 64 (Subgroups of Cyclic Groups)
Let $G = \langle x \rangle$ be cyclic.

- Every subgroup of G is cyclic. If $H \leq G$, then either $H = \{1\}$ or

$$H = \langle x^d \rangle,$$

where d is the smallest positive integer such that $x^d \in H$.

- If $|G| = \infty$, then for distinct nonnegative integers $a \neq b$,

$$\langle x^a \rangle \neq \langle x^b \rangle.$$

Moreover, for every integer m ,

$$\langle x^m \rangle = \langle x^{|m|} \rangle.$$

- If $|G| = n < \infty$, then for each positive integer $a \mid n$ there is a unique subgroup of G of order a , namely

$$\langle x^{n/a} \rangle.$$

Furthermore, for every integer m ,

$$\langle x^m \rangle = \langle x^{(n,m)} \rangle,$$

so subgroups of G correspond bijectively to the positive divisors of n .

Def. 65 (Cosets)
Let $H \leq G$ and $g \in G$. Define the left and right cosets:

$$gH = \{gh : h \in H\}, \quad Hg = \{hg : h \in H\}.$$

Any element of a coset is a representative. If G is abelian then $gH = Hg$.

Prop. 66 (Coset Properties)
Let $H \leq G$ and $a, b \in G$.

- $aH = bH$ iff $b^{-1}a \in H$. In particular, $aH = H$ iff $a \in H$.
- If $aH \cap bH \neq \emptyset$, then $aH = bH$.
- $|aH| = |H|$ for all $a \in G$.

Def. 67 (Index)
The index of H in G , denoted $[G : H]$, is the number of left cosets of H in G . If $a_1H, \dots, a_tH$ are all distinct left cosets, then

$$G = a_1H \sqcup a_2H \sqcup \dots \sqcup a_tH$$

(disjoint union).

Thm. 68 (Lagrange's Theorem)

If H is a subgroup of a finite group G , then $|H| \mid |G|$. Moreover,

$$|G| = [G : H] |H|.$$

Cor. 69
If G is finite and $a \in G$, then $o(a) \mid |G|$.

Cor. 70
If G is finite, then $a^{|G|} = 1$ for all $a \in G$.

Cor. 71
If p is prime, then every group G of order p is cyclic.

Cor. 72 (Fermat's Little Theorem)
If p is prime and $a \in \mathbb{Z}/p\mathbb{Z}$, then

$$a^p \equiv a \pmod{p}.$$

Def. 73 (Euler φ -Function)
 $\varphi(n)$ is the number of positive integers $a \leq n$ with $(a, n) = 1$. If

$$n = p_1^{n_1} p_2^{n_2} \cdots p_r^{n_r} \quad (n_i > 0),$$

then

$$\varphi(n) = n(1 - p_1^{-1})(1 - p_2^{-1}) \cdots (1 - p_r^{-1}).$$

Cor. 74 (Euler's Theorem)

If $(r, m) = 1$, then

$$r^{\varphi(m)} \equiv 1 \pmod{m}.$$

Def. 75 (Conjugate)

If G is a group and $a \in G$, a conjugate of a is any element of the form

$$gag^{-1} \quad (g \in G).$$

Def. 76 (Normal Subgroup)

A subgroup $H \leq G$ is normal if

$$gHg^{-1} = H \quad \text{for all } g \in G,$$

equivalently $N_G(H) = G$. Write $H \trianglelefteq G$.

Rem. 77
Examples:

- If G is abelian, every subgroup is normal.
- $\{e\} \trianglelefteq G$, $Z(G) \trianglelefteq G$, and $G \trianglelefteq G$.
- $\langle (1, 2, 3) \rangle \trianglelefteq S_3$ and $\langle (1, 2) \rangle$ is not normal.
- $\langle r \rangle \trianglelefteq D_{2n}$.

Prop. 78 (Characterizations of Normality)

Let $H \leq G$. The following are equivalent:

- $H \trianglelefteq G$.
- $gH = Hg$ for all $g \in G$.
- $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$.

Def. 79 (Product of Subsets)

If $X, Y \subseteq G$, define

$$XY = \{xy : x \in X, y \in Y\}.$$

This multiplication is associative: $X(YZ) = (XY)Z$.

Prop. 80

Let $H, K \leq G$.

- If one of H, K is normal in G , then HK is a subgroup of G ; moreover $HK = KH$.
- If both H and K are normal in G , then $HK \trianglelefteq G$.

Thm. 81 (Quotient Group)

Let G/H be the set of left cosets of $H \leq G$. If $H \trianglelefteq G$, then

$$(aH)(bH) = abH \quad \text{for all } a, b \in G,$$

and G/H is a group under this operation.

Def. 82 (Quotient Group)

If $H \trianglelefteq G$, the group G/H is called the quotient group of G mod H .

Rem. 83

Examples:

- $G = \mathbb{Z}$, $H = \langle n \rangle$ gives $G/H = \mathbb{Z}/n\mathbb{Z}$.
- $H = \{1\}$ or $H = G$ gives $G/H = G$ or $G/H = \{e_G\}$ respectively.
- $G = \mathbb{R}$, $H = \mathbb{Z}$ gives $\mathbb{R}/\mathbb{Z}$.
- $G = \mathbb{Q}$, $H = \mathbb{Z}$ gives $\mathbb{Q}/\mathbb{Z}$.

Def. 84 (Simple Group)

A group G is simple if its only normal subgroups are $\{e\}$ and G .

Rem. 85

Examples:

- If G is abelian, G is simple iff G is cyclic of prime order p .
- A_n is simple for $n \geq 5$.

Def. 86 (Homomorphism / Isomorphism)

Let $(G, *)$ and $(H, \circ)$ be groups. A function $f : G \rightarrow H$ is a homomorphism if

$$f(x * y) = f(x) \circ f(y) \quad \text{for all } x, y \in G.$$

If f is bijective, it is an isomorphism. Write $G \cong H$ if there exists an isomorphism $G \rightarrow H$.

Def. 87 (Kernel and Image)

If $f : G \rightarrow H$ is a homomorphism, define

$$\ker f = \{x \in G : f(x) = 1\} = f^{-1}(e_H),$$

$$\operatorname{im} f = \{h \in H : h = f(x) \text{ for some } x \in G\} = f(G).$$

Lem. 88

Let $f : G \rightarrow H$ be a homomorphism.

- $f(1) = 1$.
- $f(x^{-1}) = f(x)^{-1}$. Moreover $f(x^n) = f(x)^n$ for $n \in \mathbb{Z}$.
- $\ker f \trianglelefteq G$ and $\operatorname{im} f \leq H$.

Prop. 89

Every normal subgroup $K \trianglelefteq G$ is the kernel of some homomorphism.

Thm. 90 (First Isomorphism Theorem)

If $f : G \rightarrow H$ is a homomorphism, then

$$\ker f \trianglelefteq G \quad \text{and} \quad G/\ker f \cong \operatorname{im} f.$$

More precisely, if $K = \ker f$ and $\varphi : G/K \rightarrow \operatorname{im} f$ is given by

$$\varphi(aK) = f(a),$$

then φ is an isomorphism.

Thm. 91 (Second Isomorphism Theorem)

If $H, K \leq G$ with $H \trianglelefteq G$, then $HK \leq G$, $H \cap K \trianglelefteq K$, and

$$K/(H \cap K) \cong HK/H.$$

Thm. 92 (Third Isomorphism Theorem)

If $H, K \trianglelefteq G$ with $K \leq H$, then $H/K \trianglelefteq G/K$ and

$$(G/K)/(H/K) \cong G/H.$$

Thm. 93 (Correspondence Theorem)

Let $K \trianglelefteq G$ and $\pi : G \rightarrow G/K$ be the natural map. Then

$$S \mapsto \pi(S) = S/K$$

is a bijection between $\operatorname{Sub}(G; K) = \{S \leq G : K \leq S\}$ and $\operatorname{Sub}(G/K)$. Moreover, writing $S^* = S/K$:

- $K \leq T \leq S \leq G \iff T^* \leq S^*$, and then $[S : T] = [S^* : T^*]$;
- $T \trianglelefteq S \iff T^* \trianglelefteq S^*$, and then $S/T \cong S^*/T^*$.

Prop. 94

If G is a finite abelian group and $d \mid |G|$, then G contains a subgroup of order d .

Def. 95 (Automorphism)

An isomorphism from G to itself is an automorphism. The set of all automorphisms forms a group under composition, denoted $\operatorname{Aut}(G)$.

Rem. 96

Examples:

- $\operatorname{Aut}(\mathbb{Z}/n\mathbb{Z}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.
- For $g \in G$, define $c_g : G \rightarrow G$ by $c_g(x) = gxg^{-1}$. Then $c_g \in \operatorname{Aut}(G)$ and $g \mapsto c_g$ is a homomorphism $G \rightarrow \operatorname{Aut}(G)$.

Def. 97 (Inner Automorphism)

For $g \in G$, c_g is called an inner automorphism. The subgroup of $\operatorname{Aut}(G)$ consisting of all inner automorphisms is denoted $\operatorname{Inn}(G)$.

Rem. 98

Fact: $\operatorname{Inn}(G) \cong G/Z(G)$.

Example: $\operatorname{Aut}(V_4) \cong S_3$ and $\operatorname{Inn}(V_4) = \{1\}$.

Def. 99 (Direct Product)

If H and K are groups, their direct product $H \times K$ is the set of pairs (h, k) with operation

$$(h, k)(h', k') = (hh', kk').$$

Rem. 100

Examples:

- $V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.
- If $(m, n) = 1$, then $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ (Chinese Remainder Theorem special case).

Prop. 101

If G contains normal subgroups H and K with

$$H \cap K = \{1\} \quad \text{and} \quad HK = G,$$

then $G \cong H \times K$.

Rem. 102

The expression $hkh^{-1}k^{-1}$ is called the commutator of h and k .

Rem. 103

If $G = G_1 \times \cdots \times G_n$, then $G_i \trianglelefteq G$ and

$$G/G_i \cong G_1 \times \cdots \times G_{i-1} \times G_{i+1} \times \cdots \times G_n.$$

Thm. 104 (Jordan–Hölder Theorem)

Let G be a finite group with $G \neq 1$.

- G has a composition series: $1 = N_0 \trianglelefteq N_1 \trianglelefteq \cdots \trianglelefteq N_r = G$, with $N_{i-1} \trianglelefteq N_i$ and N_i/N_{i-1} simple.
- Composition factors are unique up to order: if $1 = N_0 \trianglelefteq \cdots \trianglelefteq N_r = G$ and $1 = M_0 \trianglelefteq \cdots \trianglelefteq M_s = G$ are composition series, then $r = s$ and there is a permutation π of $\{1, \dots, r\}$ such that

$$M_{\pi(i)}/M_{\pi(i)-1} \cong N_i/N_{i-1} \quad (1 \leq i \leq r).$$

Thm. 105 (Cayley's Theorem)

Every group G is isomorphic to a subgroup of the symmetric group S_G . In particular, if $|G| = n$, then G is isomorphic to a subgroup of S_n .

Def. 106 (Group Action / G-set)

Let X be a set and G a group. We say G acts on X if there is a function $G \times X \rightarrow X$, written $(g, x) \mapsto gx$, such that:

- $(gh)x = g(hx)$ for all $g, h \in G$, $x \in X$;
- $1x = x$ for all $x \in X$.

In this case X is a G -set.

Def. 107 (Orbit and Stabilizer)

If G acts on X and $x \in X$, define the orbit

$$O(x) = \{gx : g \in G\} \subseteq X,$$

and the stabilizer

$$G_x = \{g \in G : gx = x\} \leq G.$$

Rem. 108

Define $x \sim y$ iff $\exists g \in G$ with $y = gx$. This is an equivalence relation; its equivalence classes are the orbits. G acts transitively on X iff there is only one orbit.

Rem. 109 (Standard Actions)

Examples:

- (Conjugation action) G acts on itself via $c_g(x) = gxg^{-1}$.
- If $H \leq G$, then G acts on G/H by left multiplication: $\tau_g(aH) = gaH$. The map $g \mapsto \tau_g$ is a homomorphism $G \rightarrow S_{G/H}$.

Prop. 110

Let $H \leq G$. Let G act on G/H by left multiplication, and let $\pi_H : G \rightarrow S_{G/H}$ be the homomorphism $g \mapsto \tau_g$. Then:

- G acts transitively on G/H .
- The stabilizer of $aH \in G/H$ is aHa^{-1} .
- $\ker \pi_H = \bigcap_{x \in G} xHx^{-1}$, and $\ker \pi_H$ is the largest normal subgroup of G contained in H .

Prop. 111 (Orbit Decomposition)

If G acts on a set X , then X is the disjoint union of its orbits. If X is finite, then

$$|X| = \sum_i |O(x_i)|,$$

where x_i is chosen from each orbit.

Thm. 112 (Orbit–Stabilizer)

If G acts on X and $x \in X$, then

$$|O(x)| = [G : G_x].$$

Rem. 113 (Class Equation)
If G is finite, then

$$|G| = |Z(G)| + \sum_i |G : C_G(x_i)|,$$

where x_i is chosen from each conjugacy class of size > 1 .

Thm. 114 (Cauchy’s Theorem)

If G is a finite group and a prime p divides $|G|$, then G contains an element of order p .

Def. 115 (p -group)

Let p be prime. A finite group G is a p -group if $|G| = p^n$ for some $n \geq 0$. An infinite group G is a p -group if every element has order a power of p .

Thm. 116

Let p be prime and P be a p -group.

1. $Z(P) \neq 1$.
2. If $H \trianglelefteq P$ is nontrivial, then $H \cap Z(P) \neq 1$. In particular, every normal subgroup of order p is contained in $Z(P)$.

Cor. 117

Let P be a p -group with $|P| = p^k$. Then P has a normal subgroup of order p^m for every $1 \leq m \leq k$.

Def. 118 (Sylow p -subgroup)

Let G be a group and p a prime.

1. A p -subgroup is a subgroup of G which is a p -group.
2. If $|G| = p^\alpha m$ with $p \nmid m$, a subgroup of order p^α is a Sylow p -subgroup.
3. Let $\text{Syl}_p(G)$ be the set of Sylow p -subgroups, and let $n_p(G) = |\text{Syl}_p(G)|$.

Thm. 119 (The Sylow Theorems)

Let $|G| = p^\alpha m$ with $p \nmid m$.

1. $\text{Syl}_p(G) \neq \emptyset$.
2. If $P \in \text{Syl}_p(G)$ and Q is any p -subgroup, then $\exists g \in G$ such that $Q \leq gPg^{-1}$. In particular, any two Sylow p -subgroups are conjugate.
3. $n_p \equiv 1 \pmod{p}$. Moreover, for any Sylow p -subgroup P , $n_p = [G : N_G(P)]$, hence $n_p \mid m$.

Lem. 120

Let $P \in \text{Syl}_p(G)$. If Q is any p -subgroup of G , then

$$Q \cap N_G(P) = Q \cap P.$$

Cor. 121

Let P be a Sylow p -subgroup of G . The following are equivalent:

1. P is the unique Sylow p -subgroup ($n_p = 1$).
2. $P \trianglelefteq G$.
3. If $X \subseteq G$ and $o(x)$ is a power of p for all $x \in X$, then $\langle X \rangle$ is a p -group.

Rem. 122 (Order pq Case)

Let $|G| = pq$ for primes p, q with $p \nmid (q-1)$ and $q \nmid (p-1)$. Then G is abelian (indeed cyclic).

Rem. 123

If $p \mid (q-1)$, there exists a unique non-abelian group of order pq with $n_p = q$ (e.g. $pq = 6$ gives S_3).

Rem. 124 (Techniques for Normal Subgroups)

Useful ways to produce normal subgroups:

1. **Counting elements.** Use Sylow numbers, disjointness of distinct Sylow p -subgroups, or uniqueness by order/property. If a subgroup is unique, then it is normal.
2. **Permutation representations.** Let G act on something and obtain a homomorphism

$$G \rightarrow S_n.$$

The kernel is normal. Typical choice: action on cosets G/H .

3. **Playing with p -subgroups for different primes.** Use conjugation actions

$$Q \rightarrow \text{Aut}(P) \quad \text{or} \quad P \rightarrow \text{Aut}(Q)$$

for Sylow subgroups P, Q of different prime-power orders. Divisibility often forces the image to be trivial, so the action is trivial, hence the subgroups centralize or normalize each other.

Prop. 125

Let $|G| = 30$. Then G has a normal subgroup isomorphic to $\mathbb{Z}/15\mathbb{Z}$.

Rem. 126

It is enough to produce a subgroup of order 15, since any subgroup of index 2 is normal. Such a subgroup is cyclic because $3 \nmid (5-1)$.

Let $P \in \text{Syl}_5(G)$ and $Q \in \text{Syl}_3(G)$. If either P or Q is normal, then PQ is a subgroup of order 15, and we are done.

Assume neither is normal. Then by Sylow, $n_5 = 6$, $n_3 = 10$. Write $\text{Syl}_5(G) = \{P_1, \dots, P_6\}$, $\text{Syl}_3(G) = \{Q_1, \dots, Q_{10}\}$. Distinct Sylow 5-subgroups intersect trivially, so the union $P_1 \cup \dots \cup P_6$ contains 24 non-identity elements of order 5. Likewise, the union $Q_1 \cup \dots \cup Q_{10}$ contains 20 non-identity elements of order 3.

These sets are disjoint, so G would contain at least $24 + 20 + 1 = 45$ elements, impossible since $|G| = 30$. Therefore one of P, Q is normal, and hence G has a normal subgroup of order 15, necessarily isomorphic to $\mathbb{Z}/15\mathbb{Z}$.

Lem. 127

Let G be a finite simple group and let $H < G$. Then $|G| \mid [G : H]!$. In particular, there is no simple group of order $p^\alpha m$ with $p \nmid m$ and $p^\alpha \nmid (m-1)!$.

Rem. 128

Let G act on G/H by left multiplication. This gives a homomorphism $\pi_H : G \rightarrow S_{[G:H]}$. Since H is proper, this map is not trivial. If G is simple, its kernel must be trivial, so G embeds into $S_{[G:H]}$. Therefore $|G| \mid |S_{[G:H]}| = [G : H]!$. The final statement follows immediately.

Prop. 129

There is no simple group of order 3393.

Rem. 130

Assume G is simple of order $3393 = 3^2 \cdot 13 \cdot 29$. By the lemma above, every proper subgroup has index at least 29.

Now let $P \in \text{Syl}_3(G)$. By Sylow, $n_3 \equiv 1 \pmod{3}$, $n_3 \mid 13 \cdot 29$. Hence $n_3 = 1$ or 13. Since G is simple, $n_3 \neq 1$, so $n_3 = 13$. Thus $[G : N_G(P)] = 13$, contradicting the fact that no proper subgroup can have index below 29.

Prop. 131

There is no simple group of order 396.

Rem. 132

Assume G is simple of order $396 = 2^2 \cdot 3^2 \cdot 11$. By Sylow, $n_{11} \equiv 1 \pmod{11}$, $n_{11} \mid 36$, so $n_{11} = 12$. Let $P \in \text{Syl}_{11}(G)$. Then $[G : N_G(P)] = 12$, $|N_G(P)| = \frac{396}{12} = 33$. Hence G embeds in S_{12} and contains a subgroup of order 33.

But every group of order 33 is cyclic, so such a subgroup would be generated by an element of order 33. In S_{12} , an element of order 33 would require disjoint cycles of lengths 3 and 11, which is impossible since $3 + 11 > 12$. Contradiction.

Prop. 133

There is no simple group of order 1785.

Rem. 134

Assume G is simple of order $1785 = 3 \cdot 5 \cdot 7 \cdot 17$. By Sylow, $n_{17} \equiv 1 \pmod{17}$, $n_{17} \mid 105$, so $n_{17} = 35$. Let $Q \in \text{Syl}_{17}(G)$. Then $[G : N_G(Q)] = 35$, $|N_G(Q)| = \frac{1785}{35} = 3 \cdot 17$. Let $P \in \text{Syl}_3(N_G(Q))$. Since $3 \nmid (17-1)$, the subgroup PQ is abelian. In particular, $Q \leq N_G(P)$, so $17 \mid |N_G(P)|$.

Now $P \in \text{Syl}_3(G)$, and by Sylow, $n_3 \equiv 1$

$\pmod{3}$, $n_3 \mid 5 \cdot 7 \cdot 17 = 595$. The possible values are $n_3 \in \{1, 7, 85, 595\}$. Since $17 \mid |N_G(P)|$, we have $n_3 = [G : N_G(P)]$ not divisible by 17, so $n_3 = 1$ or 7. Simplicity rules out $n_3 = 1$, hence $n_3 = 7$.

Thus G has a subgroup of index 7, contradicting the lemma that a simple group of this order cannot have a proper subgroup of index below 17.

Lem. 135 (Simple Group Index Bound)

Let G be a finite simple group. Then for every proper subgroup $H < G$,

$$|G| \mid [G : H]!$$

In particular, there is no simple group G of order $p^\alpha m$ ($p \nmid m$) such that

$$p^\alpha \nmid (m-1)!$$

Def. 136 (Finitely Generated)

A group G is finitely generated if there exists a finite subset $A \subseteq G$ such that $G = \langle A \rangle$.

Def. 137 (Free Abelian Group)

For $r \in \mathbb{Z}$, $r \geq 0$, define

$$\mathbb{Z}^r = \underbrace{\mathbb{Z} \times \dots \times \mathbb{Z}}_{r \text{ copies}}, \quad \mathbb{Z}^0 = 1.$$

Then $\mathbb{Z}^r$ is the free abelian group of rank r .

Thm. 138 (Fundamental Theorem of Finitely Generated Abelian Groups)

If G is a finitely generated abelian group, then

$$G \cong \mathbb{Z}^r \times G_0,$$

where G_0 is a finite abelian group.

Thm. 139 (Structure of Finite Abelian Groups)

Let G be an abelian group of order $n > 1$ and write

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$$

with distinct primes p_i . Then:

1. $G \cong A_1 \times A_2 \times \dots \times A_k$ where $|A_i| = p_i^{\alpha_i}$.
2. If $|A| = p^\alpha$, then

$$A \cong \mathbb{Z}_{p^{\beta_1}} \times \mathbb{Z}_{p^{\beta_2}} \times \dots \times \mathbb{Z}_{p^{\beta_t}},$$

- with $\beta_1 \geq \beta_2 \geq \dots \geq \beta_t \geq 1$ and $\beta_1 + \dots + \beta_t = \alpha$.
3. The decompositions in (1) and (2) are unique up to permutation of factors.

Def. 140 (Partition)

A partition of n is a sequence $(n_1, \dots, n_t)$ of positive integers such that

$$n_1 \geq n_2 \geq \dots \geq n_t \geq 1, \quad n_1 + \dots + n_t = n.$$

Lem. 141 (Splitting in Elementary Abelian p -groups)

Let E be an elementary abelian p -group (i.e. $x^p = 1$ for all $x \in E$). Then for any $x \in E$ there exists $M \leq E$ such that

$$E = M \times \langle x \rangle.$$

Rem. 142

An elementary abelian p -group E is isomorphic to $\mathbb{Z}_p^s$ for some $s \geq 0$.

Rem. 143

Under usual addition/multiplication, $\mathbb{Z}_p$ is a field $\mathbb{F}_p$. An elementary abelian p -group can be viewed as a vector space over $\mathbb{F}_p$.

Def. 144 (Semidirect Product)

Let H, K be groups and let $\varphi : K \rightarrow \text{Aut}(H)$ be a homomorphism. The semidirect product $H \rtimes_\varphi K$ is the set $H \times K$ with multiplication

$$(h_1, k_1)(h_2, k_2) = (h_1 \varphi(k_1)(h_2), k_1 k_2).$$

(Equivalently, $(h_1, k_1)(h_2, k_2) = (h_1(k_1 \cdot h_2), k_1 k_2)$.)

Prop. 145

$H \rtimes_\varphi K$ is a group. Its identity is $(1, 1)$ and

$$(h, k)^{-1} = (k^{-1} \cdot h^{-1}, k^{-1}).$$

If φ is trivial, then $H \rtimes_\varphi K \cong H \times K$.

Thm. 146 (Internal Semidirect Product Criterion)

Let G be a group with subgroups H, K such that:

1. $H \trianglelefteq G$,
2. $H \cap K = 1$.

Define $\varphi : K \rightarrow \text{Aut}(H)$ by conjugation:

$$\varphi(k)(h) = khk^{-1} \quad (k \in K, h \in H).$$

Then $HK \cong H \rtimes K$. In particular, if $G = HK$, then $G \cong H \rtimes K$.

Prop. 147

Let p be a prime with $p \neq 2$. Then $(\mathbb{Z}/p^n\mathbb{Z})^\times$ is cyclic.

Lem. 148 (Index p subgroup is normal)

Let G be a finite group, and let p be the smallest prime divisor of $|G|$. If $H \leq G$ and $[G : H] = p$, then H is normal in G .

Rem. 149 (Groups of Order pq)

Let $|G| = pq$ with primes $p < q$, and let $P \in \text{Syl}_p(G)$, $Q \in \text{Syl}_q(G)$. Then $n_q = 1$, so $Q \trianglelefteq G$, and $G \cong Q \rtimes P$ for some $\varphi : P \rightarrow \text{Aut}(Q)$. Also $\text{Aut}(Q) \cong (\mathbb{Z}/q\mathbb{Z})^\times$ is cyclic of order $q-1$.

- If $p \nmid (q-1)$, then φ is trivial and $G \cong Q \times P$.
- If $p \mid (q-1)$, then there is a unique non-abelian group of order pq (up to isomorphism).

Rem. 150 (Groups of Order 12)

There are precisely 5 groups of order 12 up to isomorphism:

$$\mathbb{Z}_{12}, \quad \mathbb{Z}_2 \times \mathbb{Z}_6, \quad A_4, \quad D_{12}, \quad \mathbb{Z}_3 \rtimes \mathbb{Z}_4.$$

Rem. 151

For prime p , both $(\mathbb{Z}_p, +)$ and $(\mathbb{Z}_p \setminus \{\bar{0}\}, \times)$ are cyclic. $\mathbb{Z}_p$ is the finite field $\mathbb{F}_p$. Let $V = \mathbb{F}_p^n$, an elementary abelian group under addition with $|V| = p^n$.

Prop. 152

For $V = \mathbb{F}_p^n$ under addition,

$$\text{Aut}(V) = GL_n(\mathbb{F}_p),$$

$|\text{Aut}(V)| = (p^n - 1)(p^n - p)(p^n - p^2) \dots (p^n - p^{n-1})$.

Rem. 153

$\text{Aut}(V) = GL_2(\mathbb{F}_2) \cong S_3$. Also $SL_2(\mathbb{F}_p)/\{\pm I_2\}$ is simple for primes $p \geq 5$.

Prop. 154

$GL_2(\mathbb{F}_p)$ has $p+1$ Sylow p -subgroups.

Rem. 155 (Groups of Order p^3 (p odd))

Let $|G| = p^3$ with p an odd prime.

- If G is abelian, then $G \cong \mathbb{Z}_p \times \mathbb{Z}_p \times \mathbb{Z}_p$, $\mathbb{Z}_p \times \mathbb{Z}_{p^2}$, or $\mathbb{Z}_{p^3}$.
- If G is non-abelian, then $|Z(G)| = p$ and there are exactly two non-isomorphic non-abelian groups:
 1. Case 1 (has element of order p^2):

$$G = \langle x, y \mid x^{p^2} = y^p = 1, xyx^{-1} = x^{1+p} \rangle.$$

2. Case 2 (all non-identity elements have order p):

$$G = \langle a, b, x \mid a^p = b^p = x^p = 1, ab = ba,$$

$$xax^{-1} = ab, xbx^{-1} = b \rangle.$$

In total, there are 5 non-isomorphic groups of order p^3 .

Rem. 156

For $G = S_n$, the sign map

$$\text{sgn} : S_n \rightarrow \mathbb{C}^\times$$

is a character.

Tutorial

Prop. 157 (All Bézout Solutions)
Let $a,b\in\mathbb{Z}$ with $d=(a,b)$. If (x_0,y_0) satisfies $ax_0+by_0=d$, then all integer solutions to

$$ax+by=d$$

are

$$x=x_0+\frac{b}{d}t,\qquad y=y_0-\frac{a}{d}t,\qquad t\in\mathbb{Z}.$$

Prop. 158 (Inclusion–Exclusion for φ)
If $n=p_1^{e_1}\cdots p_r^{e_r}$, then the number of integers $1\leq m\leq n$ with $(m,n)=1$ is

$$\varphi(n)=n\Bigl(1-\frac{1}{p_1}\Bigr)\Bigl(1-\frac{1}{p_2}\Bigr)\cdots\Bigl(1-\frac{1}{p_r}\Bigr).$$

Prop. 159 (Order of A_n)

$$|A_n|=\frac{n!}{2}.$$

Prop. 160
For any $\sigma\in S_n$, we have $\sigma^2\in A_n$.

Prop. 161 (Generators of S_n)
 S_n is generated by $(1,2)$ and $(1,2,3,\ldots,n)$.

Prop. 162 (Exponent 2 Implies Abelian)
If G is a group such that $x^2=e$ for all $x\in G$, then G is abelian.

Prop. 163 (Product of All Elements in Finite Abelian Group)
If G is a finite abelian group and $a=\prod_{g\in G}g$ (product over all elements), then

$$a^2=e.$$

Prop. 164 (Order-4, No Element of Order 4 $\Rightarrow V_4$)
If $|G|=4$ and G has no element of order 4, then $G\cong V_4$ (the Klein 4-group).

Prop. 165
If $a,b\in G$ satisfy $ab=ba^{-1}$ and $ba=ab^{-1}$, then

$$a^4=e\quad\text{and}\quad b^4=e.$$

Def. 166 (Torsion Subset)

$$\mathrm{Tor}(G):=\{g\in G: o(g)<\infty\}.$$

Prop. 167
If G is abelian, then $\mathrm{Tor}(G)$ is a subgroup of G .

Prop. 168 (Size of HK (Finite Case))
If H and K are finite subgroups of G , then

$$|HK|=\frac{|H||K|}{|H\cap K|}.$$

Prop. 169 (Index Multiplication Formula)
If $H\leq K\leq G$ and $[G:H]<\infty$, then

$$[G:H]=[G:K][K:H].$$

Prop. 170
If $G/Z(G)$ is cyclic, then G is abelian.

Prop. 171 (Order pq Constraint)
If $|G|=pq$ for primes p,q , then either G is abelian or $Z(G)=1$.

Prop. 172 (Coprime Normal Subgroup is Unique by Order)
Let G be finite and $K\trianglelefteq G$. Put $a=|K|$ and $b=[G:K]$. If $(a,b)=1$, then K is the unique subgroup of G of order a .

Prop. 173 (Centralizer of an n -cycle)
Let $\sigma=(1,2,3,\ldots,n)\in S_n$. Then

$$C_{S_n}(\sigma)=\langle\sigma\rangle.$$

Prop. 174
If $|G|=2k$ where k is odd, then G has a normal subgroup of order k .

Prop. 175
For $n\geq 3$, $(\mathbb{Z}/2^n\mathbb{Z})^\times$ is not cyclic.

Prop. 176
If $m\mid n$, then $\mathbb{Z}_n/\mathbb{Z}_m\cong\mathbb{Z}_{n/m}$.

Prop. 177 (CRT Special Case)
If $(m,n)=1$, then

$$\mathbb{Z}_{mn}\cong\mathbb{Z}_m\times\mathbb{Z}_n.$$

Prop. 178

$$\mathrm{Aut}(\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/4\mathbb{Z})\cong D_8.$$

Def. 179 (Commutator Subgroup)

$$G'=\langle x^{-1}y^{-1}xy:~x,y\in G\rangle=[G,G].$$

Prop. 180
 $G'\trianglelefteq G$ and G/G' is abelian.

Prop. 181 (Largest Abelian Quotient)
 G/G' is the largest abelian quotient of G : if $H\trianglelefteq G$ and G/H is abelian, then $G'\leq H$. Conversely, if $G'\leq H$, then $H\trianglelefteq G$ and G/H is abelian.

Prop. 182 (Cyclicity Criterion)
If G is an abelian group of order n having at most one cyclic subgroup of order p for each prime divisor p of n , then G is cyclic.

Thm. 183 (Groups of Order p^2)
Every group of order p^2 is abelian.

Thm. 184 (Non-Abelian Groups of Order p^3)
Let P be a non-abelian group of order p^3 . Then

$$|Z(P)|=p.$$

Moreover,

$$P/Z(P)\cong\mathbb{Z}_p\times\mathbb{Z}_p.$$

Also,

$$[P,P]=Z(P),\qquad |[P,P]|=p.$$

Lem. 185 (Commutator Powers)
If $x,y\in G$ both commute with $[x,y]$, then for all $n\geq 1$,

$$[x,y^n]=[x,y]^n.$$

Similarly, if x,y both commute with $[x,y]$, then

$$[x^n,y]=[x,y]^n.$$

Thm. 186 (Sylow p -subgroups of S_p)
Let p be prime. The number of Sylow p -subgroups of S_p is

$$n_p=(p-2)!$$

Thm. 187 (Frattini’s Argument)
Let $H\trianglelefteq G$ and let $P\in\mathrm{Syl}_p(H)$. Then

$$G=HN_G(P).$$

Hence

$$[G:H]\mid|N_G(P)|.$$

Prop. 188
If G has no subgroup of index 2 and $G\leq S_k$, then

$$G\leq A_k.$$

Prop. 189 (Odd Sylow subgroups in A_k)
If p is odd and $P\in\mathrm{Syl}_p(S_k)$, then

$$P\in\mathrm{Syl}_p(A_k),\qquad|N_{A_k}(P)|=\frac{1}{2}|N_{S_k}(P)|.$$

Thm. 190 (Automorphisms of Direct Products of Coprime Order)
If H,K are finite groups with $(|H|,|K|)=1$, then

$$\mathrm{Aut}(H\times K)\cong\mathrm{Aut}(H)\times\mathrm{Aut}(K).$$

Thm. 191 (Automorphism Group of A_4)

$$\mathrm{Aut}(A_4)\cong S_4.$$

Also, A_4 is generated by (123) and (134) .
Thm. 192 (Groups of Order p^2q)
Let p,q be primes and $|G|=p^2q$.
If $p<q$ and $p\nmid(q-1)$, then G is abelian.
If $p>q$, then G is a semidirect product

$$G\cong P\rtimes Q$$

with $P\trianglelefteq G$, where $P\in\mathrm{Syl}_p(G)$ and $Q\in\mathrm{Syl}_q(G)$.

Thm. 193 (Groups of Order p^3)
If $|G|=p^3$, then for every $x\in G$,

$$x^p\in Z(G).$$

Prop. 194 (Useful Automorphism-Group Isomorphisms)
For cyclic groups,

$$\mathrm{Aut}(\mathbb{Z}_n)\cong(\mathbb{Z}/n\mathbb{Z})^\times.$$

Also

$$\mathrm{Aut}(\mathbb{Z}_6)\cong\mathbb{Z}_2,\quad\mathrm{Aut}(\mathbb{Z}_8)\cong\mathbb{Z}_2\times\mathbb{Z}_2,\quad\mathrm{Aut}(\mathbb{Z}_{10})\cong\mathbb{Z}_4.$$

More generally, if p is prime, then

$$\mathrm{Aut}(\mathbb{Z}_p)\cong\mathbb{Z}_{p-1}.$$

Also, for odd primes p ,

$$\mathrm{Aut}(\mathbb{Z}_{p^k})\cong\mathbb{Z}_{p^{k-1}(p-1)},$$

and for $k\geq 3$,

$$\mathrm{Aut}(\mathbb{Z}_{2^k})\cong\mathbb{Z}_2\times\mathbb{Z}_{2^{k-2}}.$$

For elementary abelian p -groups,

$$\mathrm{Aut}((\mathbb{Z}_p)^n)\cong GL_n(\mathbb{F}_p).$$

In particular,

$$\mathrm{Aut}(\mathbb{Z}_p\times\mathbb{Z}_p)\cong GL_2(\mathbb{F}_p),$$

$$\mathrm{Aut}(V_4)\cong\mathrm{Aut}(\mathbb{Z}_2\times\mathbb{Z}_2)\cong GL_2(\mathbb{F}_2)\cong S_3,$$

$$\mathrm{Aut}((\mathbb{Z}_2)^3)\cong GL_3(\mathbb{F}_2)\cong PSL_2(\mathbb{F}_7).$$

Other useful small-group examples:

$$\mathrm{Aut}(\mathbb{Z}_2\times\mathbb{Z}_4)\cong D_8,\quad\mathrm{Aut}(D_8)\cong D_8,\quad\mathrm{Aut}(Q_8)\cong S_4,$$

$$\mathrm{Aut}(A_4)\cong S_4,\quad\mathrm{Aut}(S_3)\cong S_3.$$

Since $D_6\cong S_3$, also

$$\mathrm{Aut}(D_6)\cong S_3.$$

For symmetric and alternating groups,

$$\mathrm{Aut}(S_n)\cong S_n\qquad(n\neq 2,6),$$

$$\mathrm{Aut}(A_n)\cong S_n\qquad(n\geq 3,\ n\neq 6),$$

and

$$\mathrm{Out}(S_n)\cong 1\qquad(n\neq 2,6),\qquad\mathrm{Out}(S_6)\cong\mathbb{Z}_2,$$

$$\mathrm{Out}(A_n)\cong\mathbb{Z}_2\qquad(n\geq 3,\ n\neq 6).$$

For inner and outer automorphisms,

$$\mathrm{Inn}(G)\cong G/Z(G),\qquad\mathrm{Out}(G):=\mathrm{Aut}(G)/\mathrm{Inn}(G).$$

Hence

$$\mathrm{Inn}(S_3)\cong S_3,\qquad\mathrm{Inn}(Q_8)\cong Q_8/Z(Q_8)\cong V_4.$$

If G is abelian, then

$$\mathrm{Inn}(G)=1,\qquad\mathrm{Out}(G)\cong\mathrm{Aut}(G).$$

In particular,

$$\mathrm{Out}(\mathbb{Z}_n)\cong\mathrm{Aut}(\mathbb{Z}_n)\cong(\mathbb{Z}/n\mathbb{Z})^\times,$$

$$\mathrm{Out}(V_4)\cong S_3,\qquad\mathrm{Out}(Q_8)\cong S_3.$$

For dihedral groups D_{2n} ,

$$\mathrm{Aut}(D_{2n})\cong\mathbb{Z}_n\rtimes(\mathbb{Z}_n)^\times.$$

Also

$$Z(D_{2n})=\begin{cases}1,&n\text{ odd,}\\ \mathbb{Z}_2,&n\text{ even,}\end{cases}\qquad\mathrm{Inn}(D_{2n})\cong D_{2n}/Z(D_{2n}).$$

For finite fields,

$$\mathrm{Aut}(\mathbb{F}_{p^n}/\mathbb{F}_p)\cong\mathbb{Z}_n,$$

generated by the Frobenius automorphism $x\mapsto x^p$.

More generally, for finite direct products with coprime orders,

$$\mathrm{Aut}(G\times H)\cong\mathrm{Aut}(G)\times\mathrm{Aut}(H)\qquad\text{if }(|G|,|H|)=1.$$

Groups and Centers

G	$Z(G)$
C_n	C_n
V_4	V_4
S_n ($n\geq 3$)	$\{e\}$
A_n ($n\geq 4$)	$\{e\}$
D_{2n} , n odd	$\{1\}$
D_{2n} , n even	$\{1,r^{n/2}\}$
Q_8	$\{\pm 1\}$
$GL_n(F)$	$\{\lambda I_n: \lambda\in F^\times\}$
$SL_n(F)$	$\{\lambda I_n: \lambda^n=1\}$

Useful Techniques

(1) Act on something $\Rightarrow$ homomorphism. If G acts on a set Ω with $|\Omega|=n$, then

$$G\rightarrow S_n.$$

Typical choices: action on cosets G/H , conjugation on a set of subgroups, etc.

(2) Conjugation $\Rightarrow$ automorphism group. If $H\trianglelefteq G$, then conjugation gives

$$G\rightarrow\mathrm{Aut}(H),\qquad g\mapsto(h\mapsto ghg^{-1}).$$

If $|H|=p$, then $H\cong C_p$ and

$$|\mathrm{Aut}(H)|=p-1.$$

(3) Smallest prime divisor trick. If p is the smallest prime dividing $|G|$, then any divisor of $|G|$ whose prime factors are all $< p$ must be 1. In particular, if some subgroup/image has order dividing $p-1$, then it is often forced to be trivial.

(4) Compare sizes via homomorphisms. If $\varphi:G\rightarrow K$ is a homomorphism, then

$$|G|=|\ker\varphi|\,|\mathrm{im}\,\varphi|,\qquad|\mathrm{im}\,\varphi|\mid|K|.$$

Use this with Lagrange and divisibility in the target group.

(5) Prime means rigid. If $|H|=p$ is prime, then H is cyclic, with no nontrivial proper subgroups. This often forces normality, centrality, or trivial actions.

Typical patterns.

If $[G:H]=p$, act on G/H :

$$G\rightarrow S_p.$$

Then compare $|G/\ker\varphi|$ with $p!$.

If $H\trianglelefteq G$ and $|H|=p$, conjugation gives

$$G\rightarrow\mathrm{Aut}(H),$$

and since $|\mathrm{Aut}(H)|=p-1$, the image is often trivial; hence

$$ghg^{-1}=h\quad\forall g\in G,\ h\in H,$$

so

$$H\leq Z(G).$$